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BTECH
(SEM III) THEORY EXAMINATION 2025-26
DIGITAL ELECTRONICS

TIME: 3 HRS**M.MARKS: 70****Note:** Attempt all Sections. In case of any missing data; choose suitably.**SECTION A****1. Attempt all questions in brief.****02 x 7 = 14**

Q no.	Question	Marks	CO
a.	Represent the decimal number 45 in binary and octal number systems. दशमलव संख्या 45 को बाइनरी और ऑक्टल संख्या पद्धति में व्यक्त कीजिए।	02	1
b.	Differentiate between combinational and sequential circuits (any two points). कॉम्बिनेशनल और सीक्वेंशियल सर्किट में अंतर लिखिए (कोई दो बिंदु)।	02	1
c.	State the function of an encoder. एनकोडर का कार्य लिखिए।	02	2
d.	What is a multiplexer? Write any two applications. मल्टीप्लेक्सर क्या है? इसके कोई दो अनुप्रयोग लिखिए।	02	2
e.	What is a shift register? Name any two types. शिफ्ट रजिस्टर क्या है? इसके किसी दो प्रकार के नाम लिखिए।	02	3
f.	What is a hazard in digital circuits? डिजिटल सर्किट में हैज़र्ड क्या होता है?	02	5
g.	What is noise margin in logic families? लॉजिक फैमिली में नॉइज़ मार्जिन क्या होता है?	02	5

SECTION B**2. Attempt any three of the following:****07 x 3 = 21**

a.	Explain binary subtraction using 2's complement method with a suitable example. 2 का पूरक विधि का उपयोग करते हुए बाइनरी घटाव को किसी उपयुक्त उदाहरण सहित समझाइए।	07	1
b.	Design a full subtractor circuit using two half subtractors and basic gates. दो हाफ सबट्रैक्टर और बेसिक गेट्स का उपयोग करते हुए फुल सबट्रैक्टर सर्किट डिज़ाइन कीजिए।	07	2
c.	Describe the construction and working of a Johnson counter. जॉनसन काउंटर की संरचना एवं कार्य का वर्णन कीजिए।	07	3
d.	Explain the steps involved in state assignment of synchronous sequential circuits. सिंक्रोनस अनुक्रमिक सर्किट में स्टेट असाइनमेंट के चरण समझाइए।	07	4
e.	Compare TTL, ECL and CMOS logic families with respect to power dissipation, speed and noise immunity. पावर डिसिपेशन, गति और नॉइज़ इम्यूनिटी के संदर्भ में TTL, ECL और CMOS लॉजिक फैमिली की तुलना कीजिए।	07	5



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BTECH
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DIGITAL ELECTRONICS

TIME: 3 HRS

M.MARKS: 70

SECTION C

3. Attempt any one part of the following: **07 x 1 = 07**

a.	What is Excess-3 code? Convert $(57)_{10}$ to Excess-3 code and explain the advantages of Excess-3 code over BCD. एक्सेस-3 कोड क्या है? $(57)_{10}$ को एक्सेस-3 कोड में परिवर्तित कीजिए तथा BCD के मुकाबले एक्सेस-3 कोड के लाभ समझाइए।	07	1
b.	Using K-map, minimize the Boolean function and implement it using only NAND gates: $F(A,B,C,D) = \sum m(0,2,4,5,8,10,12,13)$. के-मैप का उपयोग करके निम्न बूलियन फलन को सरल कीजिए तथा केवल NAND गेट से इम्प्लीमेंट कीजिए: $F(A,B,C,D) = \sum m(0,2,4,5,8,10,12,13)$.	07	1

4. Attempt any one part of the following: **07 x 1 = 07**

a.	Explain the operation of a 4-bit priority encoder with truth table and logic diagram. ट्रुथ टेबल और लॉजिक आरेख सहित 4-बिट प्रायोरिटी एन्कोडर के कार्य को समझाइए।	07	2
b.	Design a Full Adder using two Half Adders and basic gates. दो हाफ ऐडर तथा मूल गेट्स का उपयोग करके एक फुल ऐडर डिज़ाइन कीजिए।	07	2

5. Attempt any one part of the following: **07 x 1 = 07**

a.	Explain the working of a D flip-flop using logic diagram, truth table and characteristic equation. लॉजिक आरेख, ट्रुथ टेबल तथा कैरेक्टेरिस्टिक समीकरण सहित D फ्लिप-फ्लॉप के कार्य को समझाइए।	07	3
b.	Explain Ring Counter with diagram and applications. रिंग काउंटर को आरेख तथा अनुप्रयोगों सहित समझाइए।	07	3

6. Attempt any one part of the following: **07 x 1 = 07**

a.	Explain race conditions in asynchronous sequential circuits. एसिंक्रोनस अनुक्रमिक सर्किट में रेस कंडीशन समझाइए।	07	4
b.	Explain the complete design procedure of a synchronous sequence detector using state diagram and state table. स्टेट डायग्राम और स्टेट टेबल की सहायता से किसी सिंक्रोनस सीक्वेंस डिटेक्टर की पूर्ण डिज़ाइन प्रक्रिया समझाइए।	07	4

7. Attempt any one part of the following: **07 x 1 = 07**

a.	Explain fan-in, fan-out and power dissipation in CMOS logic family. CMOS लॉजिक फैमिली में फैन-इन, फैन-आउट एवं पावर डिसिपेशन समझाइए।	07	5
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TIME: 3 HRS**M.MARKS: 70**

b.	Differentiate between PLA and PAL. Explain how a given Boolean function can be realized using a PLA. PLA और PAL के बीच अंतर लिखिए। किसी दिए गए बूलियन फलन को PLA की सहायता से कैसे महसूस किया जा सकता है, समझाइए।	07	5
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BTECH
(SEM III) THEORY EXAMINATION 2024-25
DIGITAL ELECTRONICS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 07 = 14

Q no.	Question	CO	Level
a.	What are SOP and POS forms? एसओपी और पीओएस फॉर्म क्या हैं?	1	K2
b.	Which logic gates are called universal gates? कौन से लॉजिक गेट्स को यूनिवर्सल गेट्स कहा जाता है?	1	K3
c.	What is a magnitude comparator used for? मैग्नीट्यूड कंपैरेटर का उपयोग किस लिए किया जाता है?	2	K2
d.	What is the advantage of a synchronous counter over an asynchronous counter? एसिंक्रोनस काउंटर की तुलना में सिंक्रोनस काउंटर का क्या लाभ है?	3	K2
e.	What are essential hazards? आवश्यक खतरे क्या हैं?	4	K2
f.	What is fan-out? फैन-आउट क्या है?	5	K2
g.	Which logic family consumes the least power? कौन सी लॉजिक फॅमिली सबसे कम बिजली की खपत करती है?	5	K2

SECTION B

2. Attempt any three of the following:

7 x 3 = 21

a.	Convert the following numbers to their respective bases: निम्नलिखित संख्याओं को उनके संबंधित आधारों में परिवर्तित करें: i) $(47.625)_{10}$ to binary ii) $(101101.101)_2$ to decimal iii) $(3F9)_{16}$ to decimal iv) $(543)_8$ to binary	1	K3
b.	Discuss multiplexed display systems. How are they used in practical applications? मल्टीप्लेक्स डिस्प्ले सिस्टम पर चर्चा करें। व्यावहारिक अनुप्रयोगों में इनका उपयोग कैसे किया जाता है?	2	K2
c.	Explain flip-flop conversion. How can a JK flip-flop be converted into a D flip-flop? फ्लिप-फ्लॉप रूपांतरण की व्याख्या करें। JK फ्लिप-फ्लॉप को D फ्लिप-फ्लॉप में कैसे परिवर्तित किया जा सकता है?	3	K2
d.	What is race-free state assignment? Why is it necessary in asynchronous sequential circuits? रेस-फ्री स्टेट असाइनमेंट क्या है? एसिंक्रोनस सीक्वेंशियल सर्किट में यह क्यों जरूरी है?	4	K3
e.	What is Programmable Logic Arrays (PLA)? Explain their structure and working. प्रोग्रामेबल लॉजिक ऐरेज़ (PLA) क्या है? उनकी संरचना और कार्यप्रणाली समझाइए।	5	K2



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TIME: 3 HRS

M.MARKS: 70

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

a.	Design an excess 3 to BCD code converter एक एक्सेस 3 से BCD कोड कनवर्टर डिज़ाइन करें	1	K3
b.	Discuss the Karnaugh map (K-map) method for Boolean function simplification. Simplify the following Boolean function using K-map and also draw the simplified logic circuit using basic gates. $f(A, B, C, D) = \sum m(0, 1, 5, 6, 12, 13, 14) + d(2, 4)$ बूलियन फ़ंक्शन सरलीकरण के लिए कर्नोथ मैप (K-मैप) विधि पर चर्चा करें। K-मैप का उपयोग करके निम्नलिखित बूलियन फ़ंक्शन को सरल बनाएँ और बेसिक गेट्स का उपयोग करके सरलीकृत लॉजिक सर्किट भी बनाएँ। $f(A, B, C, D) = \sum m(0, 1, 5, 6, 12, 13, 14) + d(2, 4)$	1	K3

4. Attempt any one part of the following:

07 x 1 = 07

a.	Explain the working of a BCD adder. How does it differ from a binary adder? BCD एडर की कार्यप्रणाली को समझाएँ। यह बाइनरी एडर से किस प्रकार भिन्न है?	2	K2
b.	Describe half subtractor circuits with truth tables and circuit diagrams. सत्य तालिकाओं और सर्किट आरेखों के साथ अर्ध घटाव सर्किट का वर्णन करें।	2	K2

5. Attempt any one part of the following:

07 x 1 = 07

a.	What are synchronous counters? Explain the design and working of a 4-bit synchronous counter. सिंक्रोनस काउंटर क्या हैं? 4-बिट सिंक्रोनस काउंटर के डिज़ाइन और कार्यप्रणाली की व्याख्या करें।	3	K2
b.	Explain Johnson counters with timing diagrams and applications. टाइमिंग डायग्राम और अनुप्रयोगों के साथ जॉनसन काउंटर की व्याख्या करें।	3	K2

6. Attempt any one part of the following:

07 x 1 = 07

a.	What is the significance of state assignment? List the different techniques for state assignment स्टेट असाइनमेंट का क्या महत्व है? स्टेट असाइनमेंट के लिए विभिन्न तकनीकों की सूची बनाएं	4	K2
b.	Explain the differences between synchronous and asynchronous sequential circuits with examples. उदाहरणों के साथ सिंक्रोनस और एसिंक्रोनस अनुक्रमिक सर्किट के बीच अंतर की व्याख्या करें।	4	K2

7. Attempt any one part of the following:

07 x 1 = 07

a.	Explain the structure and operation of a static RAM (SRAM) and dynamic	5	K2
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BTECH
(SEM III) THEORY EXAMINATION 2024-25
DIGITAL ELECTRONICS

TIME: 3 HRS**M.MARKS: 70**

	RAM (DRAM). स्टैटिक रैम (SRAM) और डायनेमिक रैम (DRAM) की संरचना और संचालन की व्याख्या करें।		
b.	Describe the circuit implementation and working of TTL logic gates टीटीएल लॉजिक गेट्स के सर्किट कार्यान्वयन और कार्यप्रणाली का वर्णन करें	5	K2

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BTECH
(SEM III) THEORY EXAMINATION 2023-24
DIGITAL ELECTRONICS

TIME: 3HRS

M.MARKS: 70

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

Q no.	Question	2 x 7 = 14	Marks	CO
a.	What is the difference between binary, decimal, and hexadecimal number systems?	2	1	
b.	What is a don't-care condition in Karnaugh maps?	2	1	
c.	Explain the function of a magnitude comparator.	2	2	
d.	Differentiate between synchronous and asynchronous counters.	2	3	
e.	Explain the concept of a ring counter.	2	3	
f.	What are the advantages of synchronous counters over asynchronous counters?	2	4	
g.	Differentiate between RAM and ROM.	2	5	

SECTION B

2. Attempt any three of the following:

Q no.	Question	7 x 3 = 21	Marks	CO
a.	Explain the SOP and POS forms in Boolean algebra. How are they derived?	7	1	
b.	Describe the operation of half and full adders. Provide truth tables and circuit diagrams.	7	2	
c.	Define storage elements and discuss the characteristics of latches and flip-flops.	7	3	
d.	Explain the concept of hazards in digital circuits and methods to eliminate them.	7	4	
e.	Compare and contrast different digital logic families such as DTL, DCTL, TTL, ECL, and CMOS in terms of their characteristics.	7	5	

SECTION C

3. Attempt any one part of the following:

Q no.	Question	7 x 1 = 7	Marks	CO
a.	Minimize the Boolean function $F(A, B, C, D) = \Sigma(0, 1, 2, 4, 6, 7, 9, 12, 14)$ using the Karnaugh map method.	7	1	
b.	Implement the Boolean function $F = AB + AC + BC$ using only NAND gates.	7	1	

4. Attempt any one part of the following:

Q no.	Question	7 x 1 = 7	Marks	CO
a.	Discuss the operation of BCD adders and their significance.	7	2	
b.	Implement a 4-to-1 multiplexer using basic logic gates.	7	2	

5. Attempt any one part of the following:

Q no.	Question	7 x 1 = 7	Marks	CO
a.	Explain the concept of ripple counters and synchronous counters.	7	3	
b.	Convert a JK flip-flop to a T flip-flop and demonstrate its operation with characteristic equations.	7	3	



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BTECH
(SEM III) THEORY EXAMINATION 2023-24
DIGITAL ELECTRONICS

TIME: 3HRS**M.MARKS: 70****6. Attempt any one part of the following:****7 x 1 = 7**

a.	Describe the process of state reduction and assignments in sequential circuit design.	7	4
b.	Discuss the concept of race-free state assignment and how it is achieved.	7	4

7. Attempt any one part of the following:**7 x 1 = 7**

a.	Describe the working principles of PLA and PAL and their applications.	7	5
b.	Explain the concepts of fan-out, fan-in, and noise margin in digital circuits.	7	5

B. TECH.
(SEM III) THEORY EXAMINATION 2022-23
DIGITAL ELECTRONICS

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- (a) Explain the signed binary number.
- (b) Define the term universal gates and their applications.
- (c) Elaborate the term Combinational Circuits.
- (d) Define BCD codes and convert $(A5D8)_{16}$ into BCD number.
- (e) Explain the term storage elements.
- (f) Illustrate the term sequential logic.
- (g) Define the use of clock in digital circuits.
- (h) Explain the term synchronous circuits.
- (i) Illustrate the use of logic families in digital circuits.
- (j) Elaborate the term Fan-in in digital circuits.

SECTION B

2. Attempt any three of the following: 10x3=30

- (a) Convert the following,
 - 1. $(5162)_{10} = ()_2$
 - 2. $(11011001)_2 = ()_{10}$
 - 3. $(6273)_{10} = ()_8$
 - 4. $(7860)_{10} = ()_{16}$
 - 5. $(A23B8)_{16} = ()_{10}$
- (b) Design 4:1 multiplexer using gates.
- (c) Elaborate the characteristic equations of S-R and J-K Flip-Flops.
- (d) Illustrate the State reduction technique for Digital Circuits.
- (e) Define the TTL (Transistor-Transistor-Logic) logic Family used for digital circuits.

SECTION C

3. Attempt any one part of the following: 10x1=10

- (a) Design an XOR gate by using NAND gate implementation.
- (b) Define the De-morgans theorem of Logic Simplification for SOP & POS forms.

4. Attempt any one part of the following: 10x1=10

- (a) Design a 4-bit adder circuit using gates.
- (b) Design a 3:8 Decoder circuit using gates.

5. Attempt any *one* part of the following:

10x1=10

- (a) Elaborate the working and circuit of a Serial-in-Serial-Out shift register.
- (b) Explain the working and circuit of a modulo-5 counter using gates.

6. Attempt any *one* part of the following:

10x1=10

- (a) Illustrate the working and applications of Asynchronous sequential circuits.
- (b) Explain the term, Hazard. Define different types of Hazards along with detection and reduction of Hazards.

7. Attempt any *one* part of the following:

10x1=10

- (a) Define the SRAM cell with working and circuit diagram along with applications.
- (b) Elaborate the PLA (Programmable Logic Array) along with working and applications.

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BTECH
(SEM III) THEORY EXAMINATION 2021-22
DIGITAL ELECTRONICS

Time: 3 Hours**Total Marks: 100****Notes:**

- Attempt all Sections and Assume any missing data.
- Appropriate marks are allotted to each question, answer accordingly.

SECTION-A	Attempt All of the following Questions in brief	Marks(10X2=20)	
Q1(a)	How are binary digits used to express the integer and fractional parts of a number?	1	
Q1(b)	Explain how BCD addition is carried out.	1	
Q1(c)	Implement a 4:1 multiplexer using 2:1 multiplexer.	2	
Q1(d)	Demultiplexer is decoder circuit with an additional enabling input. Do you agree with the above statement?	2	
Q1(e)	Give the difference between positive and negative edge triggering.	3	
Q1(f)	A flip-flop has 5 ns delay from the time the clock edge occurs to the time the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency the counter can operate reliably?	3	
Q1(g)	Define critical race and non-critical race.	4	
Q1(h)	What is the significance of state assignment?	4	
Q1(i)	Why is ECL logic faster than TTL?	5	
Q1(j)	Compare static RAM and dynamic RAM.	5	

SECTION-B	Attempt ANY THREE of the following Questions	Marks(3X10=30)	
Q2(a)	Realize a 3-input gate using 2-input gates for the following gates: (i) AND (ii) OR (iii) NAND (iv) NOR	1	
Q2(b)	(i) Implement a full subtractor circuit using only NAND gates. (ii) Using 4:1 multiplexers, implement the following function $F(A, B, C) = \sum m(0, 2, 3, 5, 7)$	2	
Q2(c)	Define bi-directional shift register. Draw and explain 3 bit bi-directional shift register using D flip-flop.	3	
Q2(d)	Design a primitive state diagram and state table for a circuit with two asynchronous inputs (X and Y) and one output Z. This circuit is to be designed so that if any change takes place on X and Y, Z is to change states. Assume initially that the two inputs never change simultaneously.	4	
Q2(e)	(i) Write a note on interfacing TTL with CMOS. (ii) Explain the parameters used to characterize logic families.	5	

SECTION-C	Attempt ANY ONE following Question	Marks (1X10=10)	
Q3(a)	Minimize the following using Tabular method $F(A, B, C, D, E) = \sum m(0, 1, 2, 3, 6, 7, 14, 15, 16, 19, 31)$	1	
Q3(b)	(i) Reduce the expression $f = \sum m(0, 1, 2, 3, 5, 7, 8, 9, 10, 12, 13)$ using K-maps and implement the real minimal expression using NAND logic. (ii) Design the logic circuit for a BCD to decimal decoder.	1	

SECTION-C	Attempt ANY ONE following Question	Marks (1X10=10)	
Q4(a)	Construct BCD adder using two 4-bit binary parallel adder and logic gates.	2	
Q4(b)	Explain 4-bit magnitude comparator.	2	



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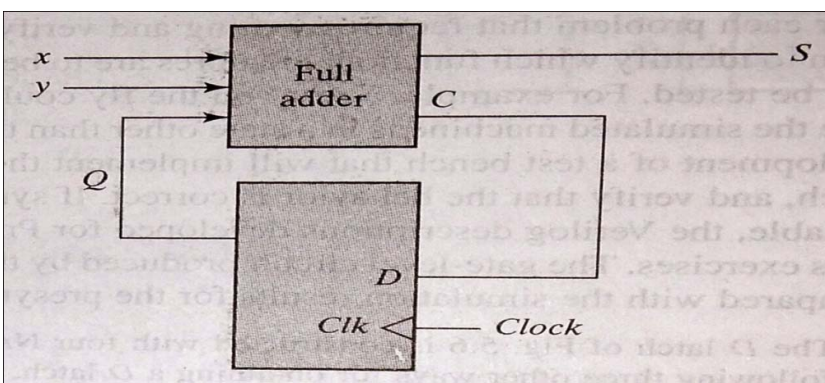
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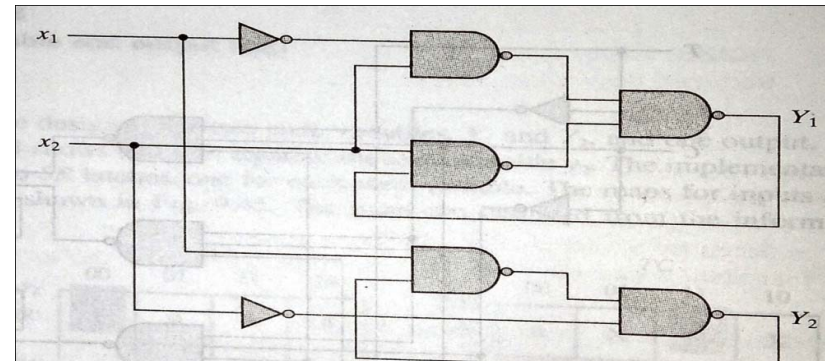
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BTECH
(SEM III) THEORY EXAMINATION 2021-22
DIGITAL ELECTRONICS

SECTION-C	Attempt ANY ONE following Question	Marks (1X10=10)	
Q5(a)	A sequential circuit has one flip-flop Q, two inputs x and y, and one output S. It consists of a full-adder circuit connected to a D flip-flop, as shown in given figure. Derive the state table and state diagram of the sequential circuit. What does this circuit represent? 		3
Q5(b)	Design a 3-bit UP/DOWN counter with a direction control M, using JK flip-flops.		3

SECTION-C	Attempt ANY ONE following Question	Marks (1X10=10)	
Q6(a)	Derive the transition table for the asynchronous sequential circuit shown in given figure. Determine the sequence of internal states Y_1Y_2 for the following sequence of inputs x_1x_2: 00,10,11,01,11,10,00. 		4
Q6(b)	(i) Distinguish between static and dynamic hazard. How will you determine hazard in combinational circuits? (ii) Draw the logic diagram of the product-of-sums expression $Y = (x_1 + x_2')(x_2 + x_3)$. Show that there is a static 0-hazard when x_1 and x_3 are equal to 0 and x_2 goes from 0 to 1. Find a way to remove the hazard by adding one more OR gate.		4

SECTION-C	Attempt ANY ONE following Question	Marks (1X10=10)	
Q7(a)	Design a BCD to Excess-3 code converter and implement it using a suitable PLA.		5
Q7(b)	Draw a neat diagram of TTL NAND gate and explain its operation.		5